Introduction

The E-Commerce Dashboard project was developed to transform raw transactional sales data into meaningful business insights through dynamic analysis and visualization in Microsoft Excel. The primary objective of this project is to create an interactive, user-friendly dashboard that enables efficient monitoring of sales and profit performance across different months, regions, and product categories.

By converting the dataset into an Excel Table and leveraging advanced functions such as **SUMIFS**, **INDEX**, and form controls like the Combo Box, the dashboard allows real-time filtering and automated updates. This ensures that decision-makers can analyze trends, compare regional performance, and evaluate category-wise profitability with accuracy and ease.

The project emphasizes scalability, automation, and clarity, demonstrating how Excel can be utilized as a powerful business intelligence tool for e-commerce analytics without requiring complex software solutions.

Step 1: Load & Prepare the Data

The original dataset is transformed into an Excel Table to enable dynamic functionality. Any changes to the table—such as adding new rows or modifying existing values—automatically update all related tables, formulas, and charts. This approach enhances accuracy, scalability, and simplifies maintenance.

Step 2: Create Month-Wise Sales & Profit Table

The month-wise total sales and total profit are calculated using Excel's SUMIFS function. This function allows us to sum values that meet multiple criteria—specifically, the selected month and the chosen product category—so we can analyze sales and profit trends for each category over time.

Steps:

1. Open E-Commerce Dashboard dataset.xlsx
2. Select the dataset → Ctrl + T → convert it into an Excel Table
    - This makes formulas, charts, and filters dynamic
3. Name the table for instance SalesData

Step 3: Create Region-Wise Sales Table

Region-wise sales performance is calculated using the SUMIFS function. Sales values are summed conditionally based on region and the selected product category, allowing clear comparison of sales across different regions.

Step 4: Create Product Category Combo Box

Add a combo box to allow users to choose a product category. The selection controls which data is displayed on the dashboard.

Purpose: Allow users to filter data dynamically

Steps:

1. Enable Developer Tab
2. File → Options → Customize Ribbon → ✓ Developer Developer → Insert → Combo Box (Form Control)
3. Draw the combo box on the dashboard
4. Create a list of Product Categories in a helper sheet
5. Right-click combo box → Format Control
    - Input Range → Product Category list
        i. Cell Link → (e.g.) B1
        ii. The linked cell will return a number representing the selected category.

Step 5: Link Combo Box to Calculations

Convert the combo box selection into a category name and use it inside Excel formulas so all summary tables update automatically when the category changes.

Goal: Change dashboard when category changes, below are the **steps**:

1. Use INDEX() to convert the combo box number into category name
   =INDEX(Category_List, B1)
2. Use this category in formulas or Pivot filters:
   a. Pivot Table → Filter → Product Category
   b. OR use SUMIFS() for dynamic tables

Step 6: Build Dynamic Charts

Create column charts based on the function-driven tables. The charts update instantly as the selected product category changes. Charts used are Combo chart, 2-D Clustered column chart.

Steps To Create Charts

3. Month-Wise Column Chart
4. Select Month-wise table
5. Insert → Column Chart
   Then Add:
   - Chart Title
   - Axis labels
   - Data labels (optional)

Region-Wise Column Chart

Repeat the same steps using the Region table

Step 7: Design the Dashboard Layout

Organize charts, tables, and controls on one worksheet with consistent formatting, clear titles, and hidden gridlines for a professional appearance.

Steps To Design the Dashboard

1. One clean worksheet named Dashboard
2. Place:
    a. Combo Box
    b. Month-wise chart
    c. Region-wise chart
    d. Use consistent colors and fonts
    e. Hide gridlines for a professional look

Conclusion

The E-Commerce Dashboard project demonstrates how structured data, conditional formulas, and interactive controls can be used to create a dynamic and scalable analytical dashboard in Excel. By enabling automatic updates and category-based filtering, the dashboard provides clear insights into monthly and regional sales performance. Overall, it showcases Excel's effectiveness as a practical tool for data analysis, reporting, and informed decision-making in e-commerce.